

The surface area of a sphere

Four times the area of a great circle — the second half of Archimedes’ result.

LESSON 47

1 × 40 MIN

GEOMETRY 11, §21

IGCSE E13.X

LESSON CLOCK

10'

12'

12'

8'

3'

The formula	10 min
Archimedes' theorem	12 min
Hemispheres and caps	12 min
Combined problems	8 min
Homework	3 min

LEARNING OBJECTIVES

- State and use $S = 4\pi R^2$.
- Relate the sphere’s area to that of its enclosing cylinder.
- Compute the curved area of a hemisphere and of a cap.
- Solve problems combining area and volume.

TEXTBOOK

National	pp. 235–240
Cambridge	Core & Extended, pp. 361–364
Workbook	Exercise 21.1

01

Explanation

The formula

$S = 4\pi R^2$

Four times the area of a great circle. A useful mental check: unroll a sphere and it covers exactly four of its own silhouettes.

R	S	V
1	4π	$\frac{4\pi}{3}$
3	36π	36π
6	144π	288π

At $R = 3$ the two numbers coincide — a coincidence of units, not a theorem. It happens because $4\pi R^2 = \frac{4}{3}\pi R^3$ exactly when $R = 3$.

Archimedes’ theorem

THE SPHERE AND ITS CYLINDER

For a sphere inscribed in a cylinder of radius R and height $2R$:

sphere : cylinder = 2 : 3 for **both** volume and total surface area

Volumes: $\frac{4}{3}\pi R^3$ against $2\pi R^3$. Areas: $4\pi R^2$ against $2\pi R^2 + 4\pi R^2 = 6\pi R^2$. Both give $\frac{2}{3}$.

More sharply still: the sphere's surface area equals the cylinder's **curved** area exactly. Any horizontal band of the sphere has the same area as the corresponding band of the cylinder — the projection stretches horizontally exactly as much as it squashes vertically. That is the theorem Archimedes asked to have carved on his tomb.

Hemispheres and caps

PART	CURVED AREA	TOTAL AREA
sphere	$4\pi R^2$	$4\pi R^2$
hemisphere	$2\pi R^2$	$3\pi R^2$
cap of height h	$2\pi Rh$	$2\pi Rh + \pi r^2$
zone between two planes h apart	$2\pi Rh$	—

A SOLID HEMISPHERE HAS A FLAT FACE

Its **curved** area is $2\pi R^2$, but a solid hemisphere also has the flat circle πR^2 , giving $3\pi R^2$. Read whether the object is a bowl or a solid dome.

The zone result is remarkable. The area of a band depends only on its height h , not on where it sits. A band near the equator and one near the pole, of equal height, have equal area.

Combined problems

Given one of S or V , the other follows through R :

$$R = \sqrt{\frac{S}{4\pi}} \quad R = \sqrt[3]{\frac{3V}{4\pi}}$$

Example. A ball has surface area 100π . Then $R = 5$ and $V = \frac{500\pi}{3} \approx 524$.

Useful relation. $\frac{V}{S} = \frac{R}{3}$ for every sphere — so a bigger ball has proportionally less surface per unit volume, which is the $\frac{1}{k}$ rule of Lesson 40 again.

02 Worked examples

EXAMPLE 1 Find the surface area of a ball of radius 7 cm.

$$1 \quad S = 4\pi(49)$$

$$2 \quad = 196\pi$$

$$3 \quad \approx 616 \text{ cm}^2.$$

ANSWER

$$196\pi \approx 616 \text{ cm}^2$$

EXAMPLE 2 A ball has surface area 144π . Find its radius and volume.

$$1 \quad 4\pi R^2 = 144\pi \Rightarrow R^2 = 36$$

$$2 \quad R = 6$$

$$3 \quad V = \frac{4}{3}\pi(216) = 288\pi \approx 905$$

ANSWER

$$R = 6, V = 288\pi$$

EXAMPLE 3 A solid hemisphere has radius 5. Find its total surface area.

$$1 \quad \text{Curved } 2\pi(25) = 50\pi.$$

$$2 \quad \text{Flat face } \pi(25) = 25\pi.$$

$$3 \quad 75\pi \approx 236.$$

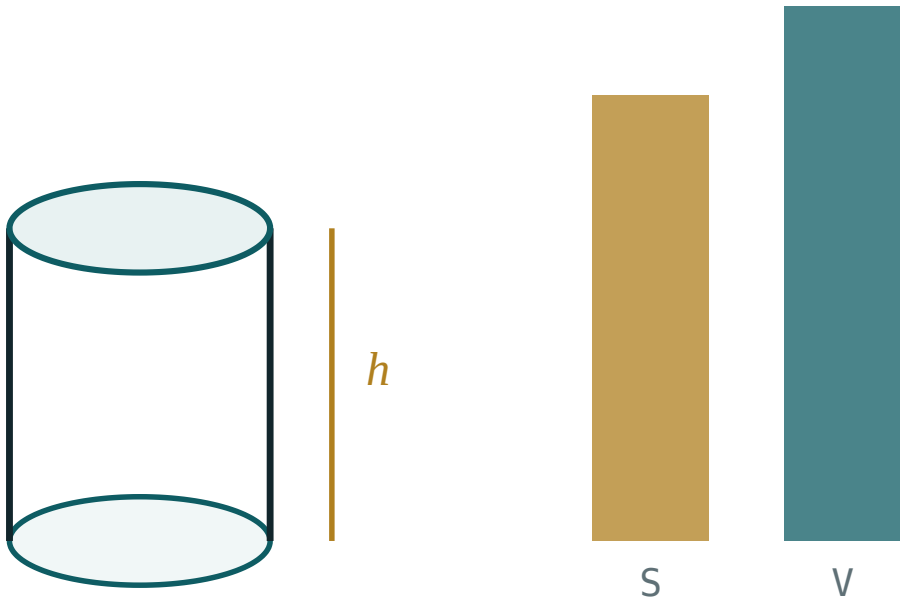
ANSWER

$$75\pi \approx 236$$

03 Interactive model

Wrap string round a ball and compare with four traced great circles.

Area against volume



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Surface area of a sphere	Sfera sirti	Площадь сферы
Great circle	Katta aylana	Большая окружность
Hemisphere	Yarim shar	Полушар
Curved surface	Yon sirt	Боковая поверхность
Zone (spherical belt)	Sharsimon belbog'	Шаровой пояс
Archimedes' theorem	Arximed teoremasi	Теорема Архимеда
Enclosing cylinder	Tashqi silindr	Описанный цилиндр
Cap	Segment	Сегмент

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The surface area of a sphere is:

πR^2

$2\pi R^2$

$4\pi R^2$

$\frac{4}{3}\pi R^3$

A hemisphere's curved area:

πR^2

$2\pi R^2$

$3\pi R^2$

$4\pi R^2$

A solid hemisphere's total area:

$2\pi R^2$

$3\pi R^2$

$4\pi R^2$

πR^2

A cap of height h has curved area:

πRh

$2\pi Rh$

$4\pi Rh$

πh^2

Sphere : cylinder for area is:

$1 : 2$

$2 : 3$

$1 : 3$

$3 : 4$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area, $R = 3$

ANSWER 36π

2. Area, $R = 5$

ANSWER 100π

3. Area, $R = 7$

ANSWER 196π

4. Area of a ball of diameter 10

ANSWER 100π

5. Hemisphere curved area, $R = 4$

ANSWER 32π

6. Solid hemisphere total area, $R = 4$

ANSWER 48π

7. Cap curved area, $R = 10, h = 3$

ANSWER 60π

Medium · the standard question

7 PROBLEMS

1. Area 144π : the radius and volume

ANSWER $6, 288\pi$

2. Solid hemisphere $R = 5$: total area

ANSWER 75π

3. A ball of volume 972π : its surface area

ANSWER 324π

4. A zone of height 4 on a sphere of radius 9

ANSWER 72π

5. Sphere : cylinder area ratio

ANSWER $2 : 3$

6. $\frac{V}{S}$ for a sphere of radius 12

ANSWER 4

7. Two balls of radii 2 and 6: the area ratio

ANSWER $1 : 9$

Hard · stretch and reason

7 PROBLEMS

1. A ball of area S : its volume in terms of S

ANSWER $\frac{S}{3} \sqrt{\frac{S}{4\pi}}$

2. A hemispherical bowl of inner radius 9 and outer 10: the material volume

ANSWER $\frac{2\pi}{3} (1000 - 729) \approx 568$

3. A sphere and a cube have equal surface areas. Which has the greater volume?

ANSWER The sphere

4. A cap of curved area 60π on a sphere of radius 10: its height

ANSWER 3

5. Prove the zone area depends only on h

ANSWER Archimedes' projection is area-preserving

6. A ball is painted at 2 so'm per cm^2 . Cost for $R = 20$

ANSWER $3200\pi \approx 10\,050$ so'm

7. A sphere of radius R inside a cube of edge $2R$: the area ratio

ANSWER $\pi : 6$

07 Homework

Homework — 4 tasks

Say whether the object is hollow or solid before choosing the formula.

- Find the surface area of a ball of radius 11 cm and of one of diameter 14 cm.
- A ball has surface area $324\pi \text{ cm}^2$. Find its radius and volume.
- Find the total surface area of a solid hemisphere of radius 8 cm.
- State Archimedes' theorem and verify both ratios for $R = 5$.